

**MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS**  
**NANYANG TECHNOLOGICAL UNIVERSITY**

Tutorial 1

HINTS

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We will discuss several questions from Chapter 1 in Ross' book.

**Tutorial Questions**

- (1) (a) How many different 7-place license plates are possible if the first 2 places are for letters and the other 5 for numbers?  
(b) Repeat part (a) under the assumption that no letter or number can be repeated in a single license plate.

**Answer:** This is a question which we can apply the multiplication rule. The difference between (a) and (b) is whether repetition is allowed.

- (a)  $26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 67\,600\,000$ .  
(b)  $26 \cdot 25 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 19\,656\,000$ .

- (2) For years, telephone area codes in the United States and Canada consisted of a sequence of three digits. The first digit was an integer between 2 and 9, the second digit was either 0 or 1, and the third digit was any integer from 1 to 9. How many area codes were possible? How many area codes starting with a 4 were possible?

**Answer:** Apply multiplication principle here again.

$$8 \cdot 2 \cdot 9 = 144; 1 \cdot 2 \cdot 9 = 18.$$

- (3) How many different letter arrangements can be made from the letters  
(a) m i s s i s s i p p i  
(b) a r r a n g e

**Answer:** (a)  $\frac{11!}{4!4!2!1!}$ ; (b)  $\frac{7!}{2!2!1!1!1!}$ .

It is an application of the multiplication principle with repetition allowed.

In (a), we have 1 m, 4 s, 4 i, 2 p, and we can have 11 positions to put 4 s, then we have 7 remaining positions, 4 for 4 i, and then we have 3 positions left, 2 for 2 i, and the only one left if for m. We then have the following number of different arrangements of these 11 letters:

$$\binom{11}{4} \binom{7}{4} \binom{3}{2} \binom{1}{1} = \frac{11!}{4!4!2!1!}.$$

For (b), the answer is  $\frac{7!}{2!2!1!1!1!}$ .

- (4) If 12 people are to be divided into 3 committees of respective sizes 3, 4, and 5, how many divisions are possible?

**Answer:**  $\frac{12!}{3!4!5!}$ . In one committee, we do not differentiate members, and we have  $\binom{12}{3}$  choices for committee I, in the remaining 9 people, we have  $\binom{9}{4}$  choices for committee II, and the remaining 5 people for committee III. By the multiplication rule, we have  $\binom{12}{3}\binom{9}{4}\binom{5}{5} = \frac{12!}{3!4!5!}$  many choices for forming these three committees.

- (5) Delegates from 10 countries, including Russia, France, England, and the United States, are to be seated in a row. How many different seating arrangements are possible if the French and English delegates are to be seated next to each other and the Russian and U.S. delegates are not to be next to each other?

**Answer:** There are several ways to solve this problem.

Method 1: We first fuse France and England as one country  $EF$ . Then we have 9 countries, and delegates from Russian and US cannot sit together. So among  $9!$  ways of seating arrangements, we need to exclude those with Russian and US delegates sitting together. This number is

$$9! - 8!2!.$$

This is for fused country  $EF$ , and now we differentiate France and England, and we have two ways: England-France, or France-England. Thus the total number of seating arrangements under the given constraints is

$$(9! - 8!2!)2!.$$

Method 2: We again fuse France and England as one country  $EF$ . Then except for Russian and US, we list  $EF$  and the other 6 countries in a row, creating 8 positions and we have  $7!$  arrangements. Note that each arrangement creates 8 positions, two at the ends, and 6 between two adjacent countries. Now we put Russian delegates and US delegates in any two such positions, we have  ${}_8P_2$  arrangements, and we have

$$7! \cdot {}_8P_2$$

arrangements of these 9 countries, with Russian and US delegates not sitting together. Now differentiate France and England, we have the number as

$$7! \cdot {}_8P_2 \cdot 2!.$$

Two methods work and the answers should be the same:

$$(9! - 8!2!)2! = 7 \cdot 8! \cdot 2!,$$

and

$$7! \cdot {}_8P_2 \cdot 2! = 7! \cdot 8 \cdot 7 \cdot 2! = 7 \cdot 8! \cdot 2! = 564\,480.$$

- (6) From a group of  $n$  people, suppose that we want to choose a committee of  $k$ ,  $k \leq n$ , one of whom is to be designated as chairperson.
- By focusing first on the choice of the committee and then on the choice of the chair, argue that there are  $\binom{n}{k}k$  possible choices.
  - By focusing first on the choice of the nonchair committee members and then on the choice of the chair, argue that there are  $\binom{n}{k-1}(n-k+1)$  possible choices.
  - By focusing first on the choice of the chair and then on the choice of the other committee members, argue that there are  $n\binom{n-1}{k-1}$  possible choices.
  - Conclude from parts (a), (b), and (c) that

$$k\binom{n}{k} = (n-k+1)\binom{n}{k-1} = n\binom{n-1}{k-1}.$$

**Answer:** (a) There are  $\binom{n}{k}$  ways to have the committee of  $n$  people and among the selected  $k$  committee members, there are  $\binom{k}{1}$  ways to choose one chairperson. In total, there are  $\binom{n}{k}\binom{k}{1} = k\binom{n}{k}$  many ways to have a committee with  $k$  members, one of whom is selected as chairperson.

(b) There are  $\binom{n}{k-1}$  ways to choose  $k-1$  nonchair members, followed by  $\binom{n-k+1}{1}$  ways of choosing chairperson. In total, there are  $\binom{n}{k-1}\binom{n-k+1}{1} = (n-k+1)\binom{n}{k-1}$  many ways to have a committee with  $k-1$  nonchair members, and one chairperson.

(c) There are  $\binom{n}{1}$  ways choose chairperson, followed by  $\binom{n-1}{k-1}$  ways of choosing non-chair members. In total, there are  $\binom{n}{1}\binom{n-1}{k-1} = n\binom{n-1}{k-1}$  many ways to have a committee one chairperson with  $k-1$  nonchair members from remaining  $n-1$  people.

(d) We obtain the equalities as all these processes give a committee of  $k$  members, with one member as chairperson.

(7) The following identity is known as Fermat's combinatorial identity:

$$\binom{n}{k} = \sum_{i=k}^n \binom{i-1}{k-1}$$

for  $n \geq k$ . Give a combinatorial argument (no computations are needed) to establish this identity.

**Answer:** Consider the number of subsets of  $A = \{1, 2, \dots, n\}$  of size  $k$ , which is  $\binom{n}{k}$ . We can also count these subsets by consider the largest number of these sets and we consider the followings cases:

- The largest number is  $k$ , then the other numbers in this set is exactly  $\{1, 2, \dots, k-1\}$ , and the number of subsets of  $A$  with  $k$  as the largest number is one, i.e.  $\binom{k-1}{k-1}$ .
- The largest number is  $k+1$ , then the other numbers ( $k-1$  many) in this set consists of  $k-1$  elements from  $\{1, 2, \dots, k\}$ , and the number of such subsets is  $\binom{k}{k-1}$ .

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- The largest number is  $n$ , then the other numbers ( $k-1$  many) in this set consists of  $k-1$  elements from  $\{1, 2, \dots, n-1\}$ , and the number of such subsets is  $\binom{n-1}{k-1}$ .

This gives

$$\binom{n}{k} = \binom{k-1}{k-1} + \binom{k}{k-1} + \dots + \binom{n-1}{k-1}.$$

- It is an example of combinatorial proof.

(8) Suppose that 10 fish are caught at a lake that contains 5 distinct types of fish.

- How many different outcomes are possible, where an outcome specifies the numbers of caught fish of each of the 5 types?
- How many outcomes are possible when 3 of the 10 fish caught are trout?
- How many when at least 2 of the 10 are trout?

**Answer:** (a) This number is the same as the number of nonnegative integer solutions to the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 10,$$

which has  $\binom{14}{4}$  many nonnegative integer solutions.

(b) Let  $x_5$  denote the number of trout in these 10 fish. This number is the same as the number of nonnegative integer solutions to the equation

$$x_1 + x_2 + x_3 + x_4 + 3 = 10,$$

i.e.

$$x_1 + x_2 + x_3 + x_4 = 7,$$

which has  $\binom{10}{3}$  many nonnegative integer solutions.

(c) Let  $x_5$  denote the number of trout in these 10 fish. Then  $x_5 \geq 2$  and we let  $x_t = y_5 + 2$ . Then the number we are calculating is the same as the number of nonnegative integer solutions to the equation

$$x_1 + x_2 + x_3 + x_4 + y_5 + 2 = 10,$$

i.e.

$$x_1 + x_2 + x_3 + x_4 + y_5 = 8,$$

which has  $\binom{12}{4}$  many nonnegative integer solutions.

- (1) How many outcome sequences are possible when a die is rolled four times, where we say, for instance, that the outcome is 3, 4, 3, 1 if the first roll landed on 3, the second on 4, the third on 3, and the fourth on 1?

**Answer:**  $6^4 = 1296$ .

- (2) A child has 12 blocks, of which 6 are black, 4 are red, 1 is white, and 1 is blue. If the child puts the blocks in a line, how many arrangements are possible?

**Answer:**  $\frac{12!}{6!4!1!1!} = 13860$ .

- (3) A student has to sell 2 books from a collection of 6 math, 7 science, and 4 economics books. How many choices are possible if  
 (a) both books are to be on the same subject?  
 (b) the books are to be on different subjects?

**Answer:** (a)  $\binom{6}{2} + \binom{7}{2} + \binom{4}{2} = 15 + 21 + 6 = 42$ ;  
 (b)  $\binom{6}{1}\binom{7}{1} + \binom{6}{1}\binom{4}{1} + \binom{7}{1}\binom{4}{1} = 94$ .

- (4) A psychology laboratory conducting dream research contains 3 rooms, with 2 beds in each room. If 3 sets of identical twins are to be assigned to these 6 beds so that each set of twins sleeps in different beds in the same room, how many assignments are possible?

**Answer:**  ${}_3P_3 \times 2 \times 2 \times 2 = 48$ . We have  ${}_3P_3$  ways to arrange three twins in 3 rooms, and in each room, there are 2 ways to arrange a twin in two different beds, and hence, there are  ${}_3P_3 \times 2 \times 2 \times 2 = 48$  ways to arrange three twins in three bedrooms, meeting the given requirements.

- (5) Ten weight lifters are competing in a team weightlifting contest. Of the lifters, 3 are from the United States, 4 are from Russia, 2 are from China, and 1 is from Canada. If the scoring takes account of the countries that the lifters represent, but not their individual identities, how many different outcomes are possible from the point of view of scores? How many different outcomes correspond to results in which the United States has 1 competitor in the top three and 2 in the bottom three?

**Answer:** (a)  $\frac{10!}{3!4!2!1!} = 12,600$ .

(b) When United States has 1 competitor in the top three and 2 in the bottom three, we have  $\binom{3}{1}$  ways to have one in the top 3 and  $\binom{3}{2}$  ways to have 2 in the bottom 3. For the other 7 competitors, and 7 positions, we have  $\frac{7!}{4!2!1!}$  different arrangements. So in total, we have  $\binom{3}{1}\binom{3}{2}\frac{7!}{4!2!1!} = 945$  arrangements to meet the given requirements.

- (6) (a) In how many ways can 3 boys and 3 girls sit in a row?  
 (b) In how many ways can 3 boys and 3 girls sit in a row if the boys and the girls are each to sit together?  
 (c) In how many ways if only the boys must sit together?  
 (d) In how many ways if no two people of the same sex are allowed to sit together?

**Answer:** (a)  $6! = 720$ .

- (b)  $3! \cdot 3! \cdot 2! = 72$ ;  
 (c) We can let 3 girls sit in a row first, and gives 4 positions for 3 boys. Thus we have  $3! \binom{4}{3} 3! = 144$ .  
 (d) We let 3 girls sit in a row first, and we have following seatings:

$B \ G \ B \ G \ B \ G$

or

$G \ B \ G \ B \ G \ B$

As no two girls can sit together, 3 boys either take the three positions on the left or the three positions on the right. Altogether, we have  $3!3!2! = 72$  ways meeting the requirement.

- (7) Give a combinatorial argument (no computations are needed) to establish the following identity:

$$\binom{n+m}{r} = \binom{n}{0} \binom{m}{r} + \binom{n}{1} \binom{m}{r-1} + \cdots + \binom{n}{r} \binom{m}{0}.$$

Then prove that

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2.$$

**Answer:** We have  $n$  boys and  $m$  girls and we will form a committee of  $r$  persons from them, so there are  $\binom{n+m}{r}$  many ways to form such a committee. We can also form such a committee by selecting  $k$  from  $n$  boys and  $r-k$  from  $m$  girls, which in total has

$$\binom{n}{0} \binom{m}{r} + \binom{n}{1} \binom{m}{r-1} + \cdots + \binom{n}{r} \binom{m}{0}$$

ways. This gives

$$\binom{n+m}{r} = \binom{n}{0} \binom{m}{r} + \binom{n}{1} \binom{m}{r-1} + \cdots + \binom{n}{r} \binom{m}{0}.$$

When  $m = n$ , take  $r = n$ , as  $\binom{n}{k} = \binom{n}{n-k}$ , we have

$$\binom{2n}{n} = \binom{n}{0} \binom{n}{n} + \binom{n}{1} \binom{n}{n-1} + \cdots + \binom{n}{n} \binom{n}{0} = \sum_{k=0}^n \binom{n}{k}^2.$$

- (8) Consider the following combinatorial identity:

$$\sum_{k=1}^n k \binom{n}{k} = n \cdot 2^{n-1}.$$

- (a) Give a combinatorial argument for this identity by considering a set of  $n$  people and determining, in two ways, the number of possible selections of a committee of any size and a chairperson for the committee.  
 (b) Give a combinatorial argument for the following identity:

$$\sum_{k=1}^n \binom{n}{k} k^2 = 2^{n-2} n(n+1).$$

- (c) By using (a) and (b), prove

$$\sum_{k=1}^n \binom{n}{k} k^3 = 2^{n-3} n^2(n+3).$$

**Answer:** (a) We consider the number of ways of selecting a committee of size  $k$ ,  $k \leq n$ , and its chairperson, which is

$$\binom{n}{k} \binom{k}{1}.$$

So the number of ways of form a committee of size  $\leq n$ , with one chairperson is

$$\sum_{k=1}^n \binom{n}{k} k.$$

Another method to get such a number is to select a chairperson first and then get other committee members from the other  $n - 1$  people, and in total, we have

$$n2^{n-1}$$

many ways. This gives the identity

$$\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}.$$

(b) We consider the number of ways of selecting a committee of size  $k$ ,  $k \leq n$ , and its chairperson, its secretary. The chairperson and its secretary can be the same person or different. So the number of such selections is

$$\sum_{k=1}^n \binom{n}{k} k \cdot k = \sum_{k=1}^n \binom{n}{k} k^2.$$

We select a chairperson first and a secretary first (note that it could be the same person), and other committee members from the remaining people, and in total, we have

$$n2^{n-1} + n(n-1)2^{n-2} = [2n + n^2 - n]2^{n-2} = n(n+1)2^{n-2}$$

many such selections. This gives the identity

$$\sum_{k=1}^n \binom{n}{k} k^2 = 2^{n-2} n(n+1).$$

(c) We consider the number of ways of selecting a committee of size  $k$ ,  $k \leq n$ , with chairperson, secretary and manager, and the number of ways of forming such a committee is

$$\sum_{k=1}^n \binom{n}{k} k \cdot k \cdot k = \sum_{k=1}^n \binom{n}{k} k^3.$$

. It could be that one person in the committee take all the roles, and the number of selections for such a committee is

$$2^{n-1}n$$

, or one person take two roles and another person takes the other role, and the number of selections for such a committee is

$$2^{n-2} \binom{n}{2} \binom{3}{2} \binom{2}{1},$$

or three persons for three positions, and number of such selections is

$$2^{n-3} \binom{n}{3} 3!.$$

In total, the number of selections for such a committee is

$$2^{n-1}n + 2^{n-2} \binom{n}{2} \binom{3}{2} \binom{2}{1} + 2^{n-3} \binom{n}{3} 3! = 2^{n-3} n^2 (n+3).$$

This gives the identity

$$\sum_{k=1}^n \binom{n}{k} k^3 = 2^{n-3} n^2 (n+3).$$

- (9) Determine the number of vectors  $(x_1, \dots, x_n)$  such that each  $x_i$  is a positive integer and

$$x_1 + x_2 + \dots + x_n \leq k$$

where  $k \geq n$ .

**Answer:** As each  $x_i$  is positive, we let  $x_i = y_i + 1$ , where  $y_i$  is nonnegative, and by  $x_1 + x_2 + \dots + x_n \leq k$ , we have

$$y_1 + y_2 + \dots + y_n \leq k - n.$$

Let  $j \leq k - n$ . Then  $y_1 + y_2 + \dots + y_n = j$  has

$$\binom{j+n-1}{n-1}$$

many nonnegative solutions, and altogether, we have

$$\binom{n-1}{n-1} + \binom{n}{n-1} + \dots + \binom{k-1}{n-1}$$

many nonnegative solutions to  $y_1 + y_2 + \dots + y_n \leq k - n$ . This gives

$$\binom{n-1}{n-1} + \binom{n}{n-1} + \dots + \binom{k-1}{n-1}$$

many positive solutions to  $x_1 + x_2 + \dots + x_n \leq k$ .

Note that

$$\begin{aligned} & \binom{n-1}{n-1} + \binom{n}{n-1} + \dots + \binom{k-1}{n-1} = \binom{n}{n} + \binom{n}{n-1} + \dots + \binom{k-1}{n-1} \\ &= \binom{n+1}{n} + \binom{n+1}{n-1} + \dots + \binom{k-1}{n-1} = \dots = \binom{k-1}{n} + \binom{k-1}{n-1} = \binom{k}{n}, \end{aligned}$$

by applying Pascal formula several times.

So there are  $\binom{k}{n}$  many such solutions.